Time-varying covariates and immortal-time bias

Inference lab using the five-step template: Hypothesis → Visualise → Assumptions → Conduct → Conclude.

Learning objectives

- Recognise immortal-time bias in a survival analysis.
- Correct it using either a landmark analysis or a time-dependent Cox model.
- Describe time-varying covariates in counting-process notation.

Prerequisites

Survival basics; `coxph` syntax.

Background

Immortal-time bias arises when a period during which the outcome cannot occur (by definition of the exposure) is misclassified as follow-up for one of the exposure groups. A classic example: "users of drug X" are defined as anyone who ever received a prescription; these people must by definition have survived until their first prescription, and if you start their follow-up at baseline rather than at the prescription, you give them "immortal" time that biases the hazard ratio toward protection.

Two standard fixes exist. Landmark analysis picks a time after baseline and classifies exposure based on what happened before that time; follow-up starts at the landmark and the bias disappears. Time-dependent Cox models treat exposure as a function of time and let patients change exposure group when they actually start the drug, using counting-process (start, stop, event) data.

Time-dependent covariates can be external (calendar time, drug availability) or internal (a lab value that evolves with the disease). Internal time-dependent covariates can introduce their own bias when conditioning on them — they may be mediators of the very effect you are studying.

Setup

```
library(tidyverse)
library(survival)
library(ggsurvfit)
set.seed(42)
theme_set(theme_minimal(base_size = 12))
```

1. Hypothesis

We simulate a cohort where every patient has the same true hazard. Some patients start a drug at a random later time. A naive analysis that counts drug users from baseline should produce a spurious protective effect.

2. Visualise

```
dat <- tibble(
  id = seq_len(n),
  t_event = rexp(n, rate = 0.02),         # true event time
```

```r
  t_drug   = rexp(n, rate = 0.05),         # time of drug start
  ever_drug = t_drug < t_event,            # did they start drug before event?
  t_drug_obs = pmin(t_drug, t_event),
  t_obs    = pmin(t_event, 100),           # admin censoring at 100
  event    = as.integer(t_event <= 100)
)

dat |>
  ggplot(aes(t_obs, fill = ever_drug)) +
  geom_histogram(bins = 30, alpha = 0.6,
                 position = "identity") +
  labs(x = "Observed time", y = "Count", fill = "Ever drug?")
```

3. Assumptions

Proportional hazards (for Cox), non-informative censoring, and independence of event times across patients. The "ever drug" split is deliberately biased by construction.

4. Conduct

```r
fit_naive <- coxph(Surv(t_obs, event) ~ ever_drug, data = dat)
summary(fit_naive)
```

```r
# classify by drug status as of 20.
landmark <- 20
dat_lm <- dat |>
  filter(t_obs > landmark) |>
  mutate(drug_lm = t_drug < landmark,
         t_obs_lm = t_obs - landmark)

fit_lm <- coxph(Surv(t_obs_lm, event) ~ drug_lm, data = dat_lm)
summary(fit_lm)
```

```r
td <- dat |>
  transmute(
    id,
    tstart = 0,
    tstop  = if_else(ever_drug, t_drug_obs, t_obs),
    event  = if_else(ever_drug, 0L, event),
    drug   = 0L
  ) |>
  bind_rows(
    dat |>
      filter(ever_drug) |>
      transmute(id,
                tstart = t_drug_obs,
                tstop  = t_obs,
                event  = event,
                drug   = 1L)
  ) |>
  filter(tstop > tstart) |>
  arrange(id, tstart)
```

```r
fit_td <- coxph(Surv(tstart, tstop, event) ~ drug, data = td)
summary(fit_td)
```

5. Concluding statement

The naive Cox analysis produced a hazard ratio of `round(exp(coef(fit_naive)), 2)` for ever-drug — spuriously protective, because immortal time was counted as exposed follow-up. A landmark analysis at time 20 gave HR `round(exp(coef(fit_lm)), 2)`, and a time-dependent Cox model gave HR `round(exp(coef(fit_td)), 2)`, both close to the null as expected from the simulated truth.

Common pitfalls

- Defining exposure over the whole follow-up based on events that occur during it.
- Picking a landmark time after visually inspecting event curves.
- Conditioning on a post-baseline mediator via a time-dependent covariate and still calling it a total effect.
- Forgetting to split the risk set at the exposure change time in counting-process data.

Further reading

- Suissa S (2008), *Immortal time bias in pharmaco-epidemiology.*
- Therneau TM, Grambsch PM. *Modeling Survival Data: Extending the Cox Model.*
- Dafni U (2011), *Landmark analysis at the 25-year landmark point.*

Session info

```
```

See also — chapter index

- Methods in this chapter
- Find your method (Chapter 0)